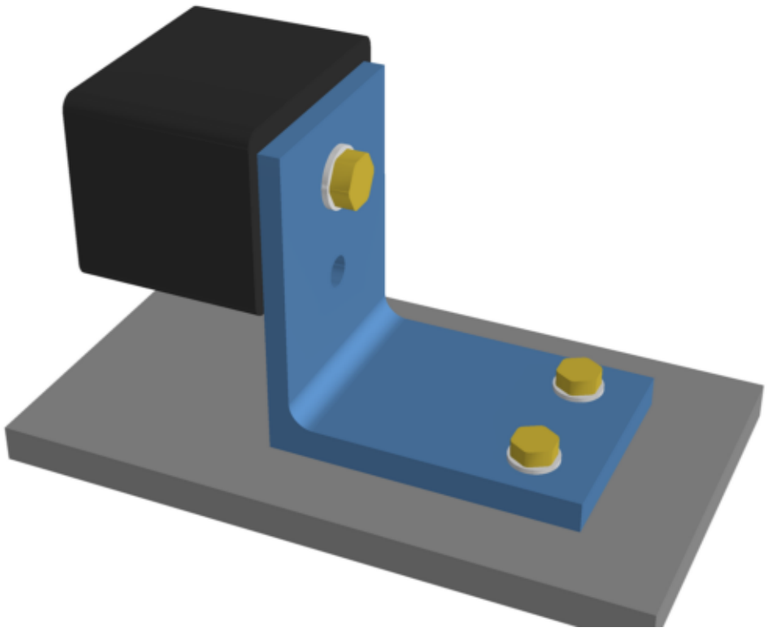
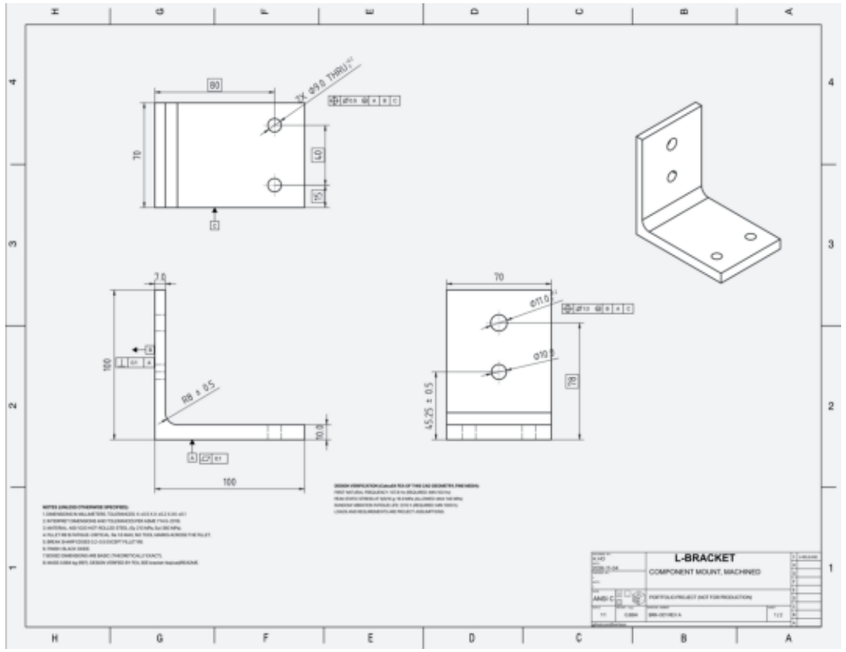
# Engine component bracket: from design study to released part

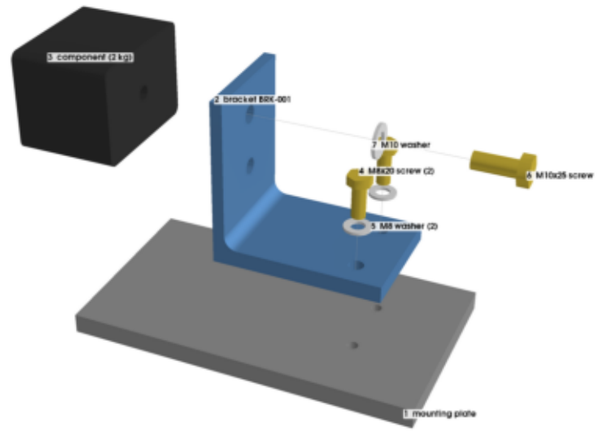
Parametric CAD (FreeCAD, Python API) · CalculiX FEA: static, modal, random vibration, fatigue · ASME Y14.5 GD&T drawings · tolerance and bolted-joint checks



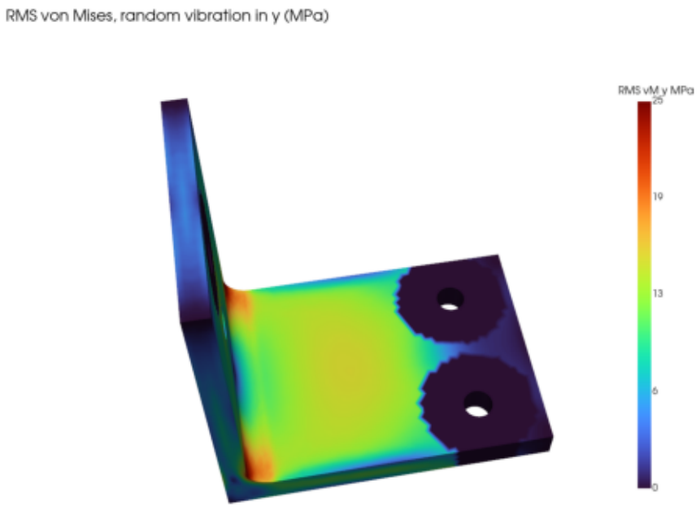
Assembly: mounting plate, bracket BRK-001, 2 kg component, ISO fasteners



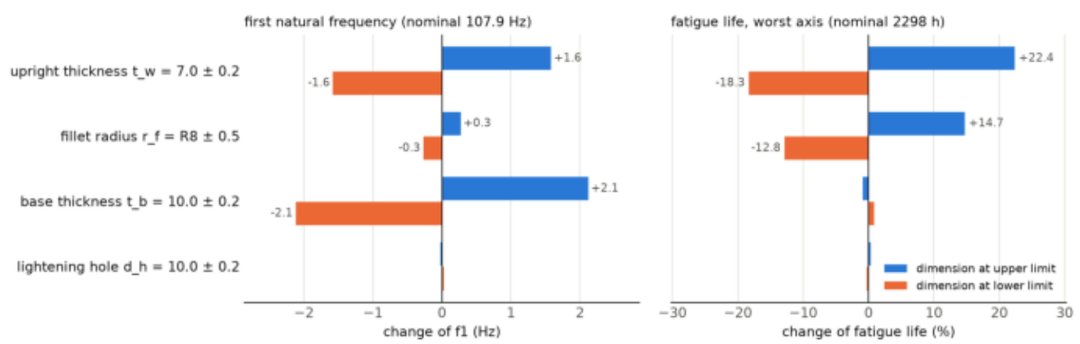

Part drawing (TechDraw, ANSI C): datums A/B/C, position at MMC, basic dimensions




Exploded view with item numbers




RMS stress, lateral random vibration (mesh from the STEP file)




Tolerance sensitivity from FEA finite differences

**Result**

| | |
|---|---|
| Mass | 0.884 kg (-5 % vs. plain bracket) |
| First natural frequency | 107.8 Hz (required ≥ 100) |
| Peak static stress | 16.9 MPa (allowed ≤ 140) |
| Fatigue life | 2,312 h (required ≥ 1000) |
| Worst-case tolerances | f1 103.9 Hz, life 1,583 h |
| CAD vs FE geometry | identical volume (2e-15) |
| Bolted joints, min SF | separation 12, slip 5.1 |

Loads and requirements are project assumptions, not a real specification.

H. Ho · github.com/RonHooo/bracket-fea